Lecture plan · for the lecturer

Evaluating models and assessing students

AI · University · Lecture 4

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Evaluate without leakage.
- Report beyond accuracy.
- Assess learning when LLMs are allowed.

Activity: Audit a project

1. Warm-up (5 min): Show a leaderboard where the winner tuned on test and ask students what the score means.
2. Teach (10 min): Show the three splits and a confusion matrix on imbalanced data.
3. Audit (20 min): Teams audit a sample project for leakage, metric choice and subgroup gaps.
4. Redesign (20 min): Rewrite one assignment so a language model cannot complete it alone.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- What leak did you find?
- Which metric would you require?
- What is your policy on LLM use?

Exit ticket

- Write one oral-defence question for your next ML project.

Try it in your next lecture
Add a ten-minute oral defence to your next project: each student explains one error their model makes and one design choice they would change.

Seminar sheet · for students

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Name _____ Class _____ Date _____

1 **Fraud is 1 in 1,000. A model flags nothing and has 99.9% accuracy. Which metric exposes it?**

◯ Accuracy ◯ Recall ◯ Training loss

2 **Work out the metrics. A model makes 40 positive predictions; 30 are right. There were 60 real positives.**

Precision = _____ %. Recall = _____ %.

Word bank: 75 · 30 · 60 · 50

3 **Which assessment designs still measure a student's own understanding when language models are allowed? Choose all that apply.**

☐ An oral defence of their notebook ☐ Error analysis on data they collected

☐ A take-home task to code a standard classifier ☐ A graded experiment log

4 **A model card should report metrics broken down by relevant subgroups.**

◯ True ◯ False

5 **Match each project flaw to the evidence that would reveal it.**

Tuned on the test set	• • • •	An oral defence
Duplicate records across splits	• • • •	Metrics disaggregated by group
Fails a subgroup	• • • •	Near-identical rows found in train and test
Student can't explain the code	• • • •	Score drops on a fresh held-out set

✎ **Which of your current ML assessments could a language model complete? What would you change first?**